Notes: Berk & Green (JPE, 2004)
Mutual Fund Flows and Performance in Rational Markets

Contents

1 Big picture

- **Question.** How can rational investors strongly chase past mutual fund performance when mutual fund performance shows little persistence and active funds do not outperform passive benchmarks on average?
- **Core answer.** The model reconciles all three facts in a competitive rational equilibrium. Investors learn about managerial ability from past returns and send more capital to high-performing managers. But because managerial skill has decreasing returns to scale, the extra capital drives expected investor alpha back to zero.
- **Main mechanism.** Good performance raises the market's posterior belief about a manager's skill. Fund inflows then expand assets under management until the marginal expected excess return to investors is again zero.
- **Why no persistence?** Past winners receive inflows that dilute their future alpha. Performance is not persistent precisely because investors rationally chase performance.
- **Why managers can be skilled even if investors earn zero alpha.** Competitive capital supply allocates money to skilled managers until investors' rents are competed away. The scarce resource is managerial ability, so rents accrue to managers through fees and scale rather than to outside investors through abnormal returns.
- **Why open-end funds are special.** In corporate finance, security prices adjust to make expected returns competitive. In open-end mutual funds, the price per share does not adjust in that way; the equilibrating margin is quantity, that is, fund flows.
- **Main quantitative result.** With a simple quadratic cost function and reasonable parameters, the model matches important empirical regularities: survival rates, a convex flow-performance relation, little performance persistence, and a high implied distribution of managerial skill.
- **Economic takeaway.** Flow-performance sensitivity is not evidence of irrationality by itself. In a rational equilibrium, performance chasing is the mechanism that makes mutual fund returns unpredictable and allocates capital to the most productive managers.

2 Incremental contribution of the paper

Incremental contribution: The paper's key contribution is to replace an apparent mutual fund puzzle with an equilibrium allocation mechanism. Before Berk and Green, three empirical facts were often treated as mutually inconsistent: active managers do not outperform benchmarks on average, fund returns show little persistence, and investors chase past performance. The paper shows that these facts can all hold simultaneously in a first-best rational market.

- **Relative to performance-persistence studies.** Earlier evidence often interpreted a lack of persistence as evidence that manager skill is nonexistent or negligible. Berk and Green show that no persistence can instead be an equilibrium consequence of rational flows that scale successful funds until their expected investor returns fall to zero.
- **Relative to behavioral explanations.** The paper does not need irrational return chasing or naive extrapolation. Investors chase performance because past returns are informative about ability; they stop earning alpha because the supply of capital is competitive.
- **Relative to agency or asymmetric-information stories.** The model has no hidden action, no hidden information between managers and investors, and no contractual distortion. The central frictions are learning and decreasing returns to scale in active management.
- **Relative to corporate finance intuition.** The paper maps mutual fund flows into the same logic as capital budgeting: investors fund positive-NPV opportunities on competitive terms, while rents accrue to the scarce input. For mutual funds, the scarce input is talent; fund flows are the quantity adjustment.

- **Quantitative contribution.** The paper shows that the flow-performance curve and survival probabilities can imply a high average skill level among active managers, not a low one. This is a sharp reinterpretation of the mutual fund evidence.

Interpretation: The incremental contribution is not merely a new formula for fund flows. It is a change in interpretation: the absence of predictable excess returns is not a failure of skill; it is the expected outcome when capital moves competitively toward skill.

3 Model objects and measurement

3.1 Managers, skill, and returns

Each mutual fund manager has an unobserved ability parameter α. Ability is the expected excess return on the first dollar the manager actively manages. For period t, the excess return on actively managed capital is written as

$$R_t = \alpha + \varepsilon_t,$$

where ε_t is idiosyncratic return noise. The variance of R_t is σ^2, and the return precision is

$$\omega = \frac{1}{\sigma^2}.$$

Interpretation: The model separates true ability from noisy realized performance. Returns are informative but imperfect signals of ability, which is why rational investors update beliefs after observing performance.

3.2 Prior beliefs and Bayesian learning

At fund birth, market participants share a common normal prior about manager ability:

$$\alpha \sim N(\phi_0, \eta^2), \qquad \gamma = \frac{1}{\eta^2}.$$

After observing a history of fund returns, investors form the posterior mean

$$\phi_t = E(\alpha \mid R_1, \dots, R_t).$$

The posterior mean is the market's current estimate of managerial skill.

Interpretation: Past returns matter because they change ϕ_t. A good return raises the posterior; a bad return lowers it. Flows are therefore rational responses to a sufficient statistic for expected ability.

3.3 Fund size, fees, and decreasing returns to scale

Let q_t denote funds under management. The total variable cost of actively managing a fund is $C(q_t)$, and the manager charges a percentage fee f. Investors bear unit costs and fees summarized by

$$c(q_t) = \frac{C(q_t)}{q_t} + f.$$

Investor return per dollar invested is

$$r_t = R_t - c(q_t).$$

The model's central technological assumption is decreasing returns to scale in active management: as a manager handles more money, the expected alpha per dollar falls because the manager's best opportunities are used first.

Interpretation: Decreasing returns are the force that prevents skilled managers from delivering persistent alpha after they receive inflows. Skill is scarce, but it is not infinitely scalable.

3.4 Competitive capital supply

Investors supply capital elastically to any fund with positive expected excess return and withdraw from funds with negative expected excess return. Therefore surviving funds satisfy

$$E_t(r_{t+1}) = 0.$$

Equivalently, fund size adjusts so that expected investor alpha is competed away.

Interpretation: This zero expected excess return condition is the key equilibrium condition. It simultaneously implies no predictable performance and a flow response to past performance.

3.5 Survival and exit

Funds also face fixed costs F. A fund survives only if the market's posterior expectation of skill is high enough to support continued operation. Let $\bar{\phi}$ denote the minimum posterior mean that allows the fund to cover fixed costs. If performance is bad enough that the posterior falls below $\bar{\phi}$, the fund exits.

Interpretation: Exit is not an ad hoc failure event. It is part of the equilibrium: poorly performing funds lose assets, cannot cover fixed costs, and shut down.

4 Models and theoretical specifications

4.1 Equilibrium condition: zero expected excess return

The equilibrium condition for surviving funds is

$$E_t(r_{t+1}) = 0.$$

Because $r_{t+1} = R_{t+1} - c(q_t)$ and $E_t(R_{t+1}) = \phi_t$, the condition implies

$$c(q_t) = \phi_t.$$

Interpretation: Fund size q_t must change when the posterior skill estimate changes. If a manager performs well, ϕ_t rises, expected returns would become positive at the old size, and investors send in capital until decreasing returns restore $c(q_t) = \phi_t$.

4.2 Bayesian updating and flow response

The paper derives a direct link between past performance and flows. A realized excess return changes posterior beliefs about ability. Because fund size is pinned down by the posterior, a return shock changes assets under management.

A stylized version of the mechanism is:

$$r_t \uparrow \quad \Rightarrow \quad \phi_t \uparrow \quad \Rightarrow \quad q_t \uparrow .$$

Interpretation: The flow-performance relationship is not imposed. It is derived from Bayesian learning and the zero expected return condition.

4.3 Quadratic cost parameterization

For the quantitative analysis, the paper assumes a quadratic cost function:

$$C(q) = aq^2.$$

Under the maintained assumptions, the optimal amount of funds under active management is

$$q^*(\phi_t) = \frac{\phi_t}{2a}.$$

The total amount of money under management is

$$q_t = \frac{\phi_t^2}{4af}.$$

The fraction of assets actively managed is

$$h(q_t) = \frac{q^*(\phi_t)}{q_t} = \sqrt{\frac{f}{aq_t}}.$$

Interpretation: A successful fund can grow very large, but the model predicts that a smaller fraction of its assets is actively managed. The rest is effectively benchmarked or passively invested. This is why growth lowers tracking error.

4.4 Flow of new funds

The model generates a convex flow-performance relationship. Higher realized returns lead to inflows, and very poor performance can lead to closure. The response is steeper for young funds because a young fund has less history, so a single return observation contains more information about ability.

$$\text{Flow response is increasing in } r_t, \quad \text{and decreasing in fund age } t.$$

Interpretation: This reproduces the empirical fact that investors react strongly to performance, especially for younger funds. The model explains this as rational learning rather than naive return chasing.

4.5 No performance persistence

Because equilibrium requires

$$E_t(r_{t+1}) = 0,$$

past performance cannot forecast future excess returns for investors. This holds even though managerial ability is heterogeneous and even though past performance is informative about ability.

Interpretation: This is the paper's central insight. Returns are not persistent because investors use performance information and move capital until future investor alpha is competed away.

4.6 Risk and past performance

The model also explains why funds with good past performance may have lower future volatility or tracking error. In successful funds, inflows exceed the scale at which active opportunities can be used productively, so the manager allocates more money to passive strategies. In poor-performing funds, outflows remove passive capital first, leaving a larger active share and higher tracking error.

Interpretation: Risk adjustment is endogenous to flows. The model can therefore explain patterns sometimes attributed to strategic risk manipulation without invoking agency problems.

5 Quantitative design and calibration strategy

5.1 What is identified in the quantitative exercise

The paper is a rational-equilibrium model with calibration, not a causal empirical design. The quantitative exercise asks whether a parsimonious model can reproduce several empirical regularities simultaneously.

- The fee f is set using evidence on mutual fund fees and loads.
- Return volatility σ is chosen to represent tracking-error and idiosyncratic volatility.
- Fixed costs and the exit threshold are linked to the minimum viable fund size.
- The mean and precision of the prior distribution of skill, (ϕ_0, γ), are chosen to match survival rates and the flow-performance relation.

Interpretation: The calibration is disciplined by multiple moments. The skill distribution is not chosen only to fit one curve; it must be consistent with survival and flows at the same time.

5.2 Why the exercise is informative

The model is useful because the same parameters must explain several facts that seem inconsistent at first glance:

1. investors chase past performance;
2. mutual fund returns do not show strong persistence;
3. active funds do not outperform passive benchmarks on average;
4. funds exit after poor histories;
5. managers can still earn large rents.

Interpretation: A behavioral story could match performance chasing alone. The contribution here is to match performance chasing while also matching no persistence and zero expected investor alpha.

5.3 Threats and limitations

- The model abstracts from moral hazard, manager career concerns, marketing, search frictions, loads, distribution networks, and fund family spillovers.
- The calibration uses an "eyeball" fit to survival rates and flow-performance curves rather than formal estimation.
- The cost function is simple and quadratic, which is useful for transparency but may not describe all active strategies.
- The model treats active-management skill as scarce and decreasing returns as central; empirical settings with hard capacity constraints or illiquidity may require richer cost functions.
- The model explains why investors do not earn persistent alpha, but it does not imply that all fees, fund structures, or investor behavior are optimal in every institutional detail.

Interpretation: The model is best interpreted as a benchmark. It tells us which mutual fund facts are not puzzling in a rational competitive market before adding agency, behavioral, or institutional frictions.

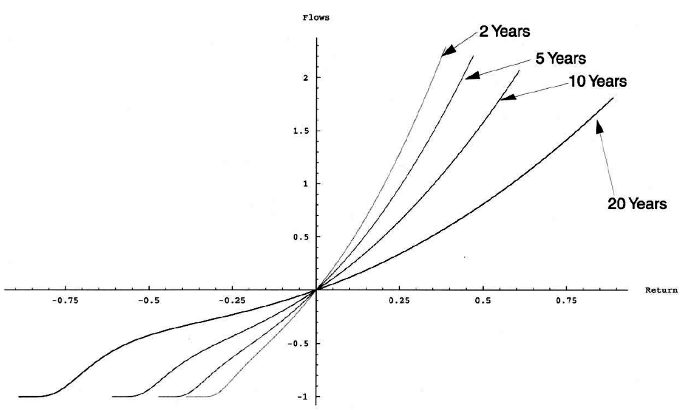
6 Tables and figures

6.1 Figure 1: Flow of New Funds as a Function of Return; Table 1: Parameter Values

Read:

- Figure 1 plots predicted new fund flows against last period's excess return for funds of different ages: two, five, 10, and 20 years.
- All curves pass through the origin: a zero excess return produces zero net new flow in the model.
- Positive returns trigger inflows and negative returns trigger outflows. For sufficiently poor returns, the lower part of the curve flattens because the fund exits and investors withdraw all capital.
- The two-year curve is steepest. Young funds have short histories, so a single return realization changes beliefs about ability more strongly.
- The 20-year curve is flatter because investors already have a lot of information; one more return observation matters less.
- Table 1 reports the parameter values used to generate the figures: $f = 1.5\%$, $\gamma = 277$, $\eta = 6\%$, $\omega = 25$, $\sigma = 20\%$, $\phi_0 = 6.5\%$, and $\bar{\phi} = 3\%$.

Economic message: Figure 1 is the cleanest visual statement of the model's learning mechanism. The same return has a bigger effect on flows when the fund is young because uncertainty about ability is greater. Table 1 shows that the curves are generated with a small set of economically interpretable parameters, not by freely fitting each curve separately.



(a) **Figure 1: Flow of new funds as a function of return.** The horizontal axis is the previous period's excess return and the vertical axis is predicted net flow. The age-specific curves show that the model predicts stronger flow-performance sensitivity for young funds and weaker sensitivity for older funds.

TABLE 1
PARAMETER VALUES

Variable	Symbol	Value
Percentage fee	f	1.5%
Prior precision	γ	277
Prior standard deviation	η	6%
Return precision	ω	25
Return standard deviation	σ	20%
Mean of prior	ϕ_0	6.5%
Exit mean	$\bar{\phi}$	3%

(b) **Table 1: Parameter Values.** These parameters discipline the model's quantitative implications. The key objects are the fee level, prior and return precision, prior mean skill, and the exit threshold.

6.2 Figure 2: Percentage of Surviving Funds

Read:

- The bars show survival rates by fund age. Light bars are the actual survival rates from CRSP mutual fund data; dark bars are the survival rates predicted by the model.
- Survival falls with age because some funds accumulate enough bad performance news that their posterior ability falls below the minimum viable threshold.
- The model predicts a geometric decline in survival rates, while the actual data decline more linearly.
- The thick line shows the number of original funds that could have survived at each age, which falls as the sample ages.

Economic message: Figure 2 tests whether the same skill distribution that explains flows can also explain fund exit. The match is not perfect, but the model captures the broad survival pattern. This matters because exit is essential for interpreting negative performance: extreme underperformance causes fund liquidation, not just a smooth outflow.

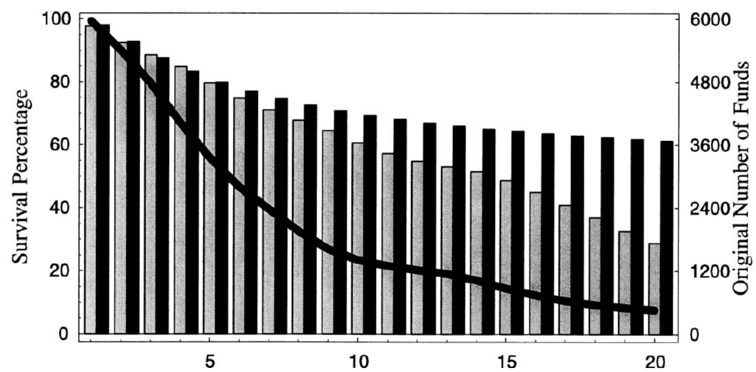


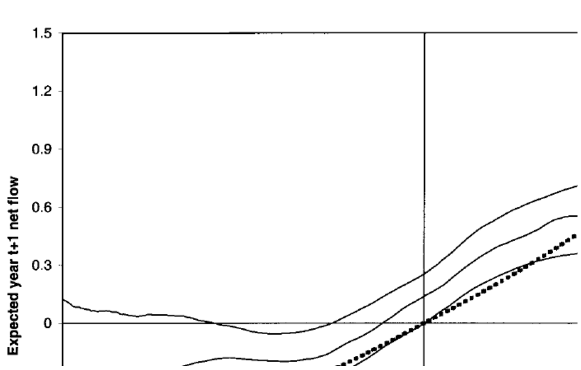
Figure 2: Percentage of surviving funds. The figure compares actual CRSP survival rates with model-predicted survival rates by fund age. The visual shows that exit is tied to performance histories and helps discipline the distribution of ability and fixed costs.

6.3 Figure 3: Flow of Funds; Figure 4: Distribution of Management Skill

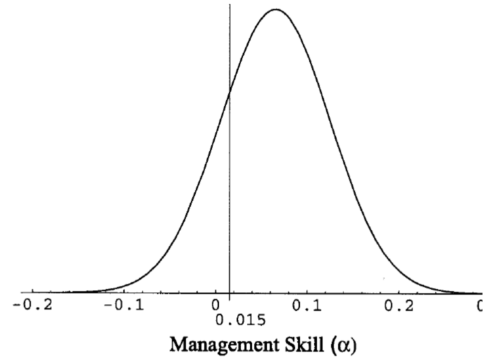
Read:

- Figure 3 overlays the model-implied flow-performance curve for two-year-old funds on the nonparametric empirical curve from Chevalier and Ellison (1997).
- The dashed model curve captures the broad positive and convex relation between returns and flows.
- The empirical curve does not pass exactly through the origin. The paper suggests that sector-wide mutual fund growth, absent from the model, can explain this discrepancy.
- Figure 4 plots the implied prior distribution of managerial skill. The vertical line marks the 1.5 percent management fee.
- Around 80 percent of the mass lies to the right of the fee line, implying that many managers can add value on the first dollar they manage.
- The average ability implied by the parameterization is about 6.5 percent.

Economic message: The two figures together show why the model's reinterpretation is powerful. A rational model can match the observed flow-performance curve while implying that managers are, on average, skilled. The absence of investor alpha is therefore not evidence of no skill; it reflects competition and scale effects.



(a) **Figure 3: Flow of funds.** The dashed model curve is superimposed on the empirical flow-performance curve and confidence bands from Chevalier and Ellison. The comparison is the paper's main quantitative validation of the flow mechanism.



(b) **Figure 4: Distribution of management skill.** The vertical line is the management fee. The mass to the right of the line measures managers whose ability exceeds fees on the first dollar.

6.4 Table A1: Important Notation

Read:

- Table A1 lists the core notation used throughout the model: realized returns, manager ability, posterior beliefs, fund size, active share, fees, costs, survival rates, and exit thresholds.
- The notation makes clear that the model distinguishes the return on the first active dollar, R_t, from the investor's net return, r_t, after costs and scale effects.
- It also distinguishes funds under management, q_t, from the amount optimally placed under active management, $q^*(\phi_t)$.

Economic message: Table A1 clarifies the central economics. Investors observe net fund returns and assets under management; from these they update beliefs about ability. Flows change q_t, while decreasing returns and passive indexing determine how much of q_t is actually actively managed.

TABLE A1
IMPORTANT NOTATION

R_t	Excess return earned on the first dollar actively managed by the fund in period t
α	Expected excess return earned on the first dollar actively managed by the fund in period t: our measure of the manager's ability
σ^2	Variance of R_t
ω	Precision of R_t: $\omega = 1/\sigma^2$
η^2	Variance of the market's prior over managerial ability, α
γ	Precision of the market's prior over managerial ability, α: $\gamma = 1/\eta^2$
ϕ_t	The market's expectation of managerial ability, conditional on the return history up to date t
ϕ_0	Mean of the market's initial prior over managerial ability
$\bar{\phi}$	The minimal expectation of managerial ability at which the fund can recover its fixed costs and continue to operate
q_t	Funds under management at date t
$q^*(\phi_t)$	Optimal amount of funds under active management, given the market's expectations of managerial ability
$\bar{q} \equiv q^*(\bar{\phi})$	The lowest value of q for which the fund remains viable
q_{It}	Funds the manager raises but passively invests in the benchmark portfolio: $q_t = q^*(\phi_t) + q_{It}$
$h(q_t)$	Proportion of funds under active management: $h(q_t) = q^*(\phi_t)/q_t$
f_t	Management fees, where the optimal level of fees is denoted f_t^*; when fees are fixed through time, $f_t = f$
F	Fixed costs of operating the fund each period
$C(q)$	The total variable costs of actively managing a fund of size q
$c(q)$	The unit costs and fees borne by investors in a fund of size q: $c(q) = [C(q)/q] + f$
r_t	Return to investors per dollar invested in the fund: $r_t = R_t - c(q_t)$
P_t	Survival rate: the probability that a fund born at date 0 survives until date t
$r^*(\phi_{t-1})$	The minimal return realization such that a fund with expected managerial ability ϕ_{t-1} survives through the next period

Table A1: Important Notation. The notation table organizes the model's state variables and choice variables. It is useful for separating ability, posterior beliefs, active-management scale, total fund size, and investor net returns.

7 Short summary

- The paper explains mutual fund flows and performance in a rational competitive equilibrium.
- Investors learn about unobserved manager ability from past returns.
- Successful funds receive inflows because good performance raises posterior beliefs about skill.
- Decreasing returns to scale in active management cause inflows to dilute future investor alpha.
- Therefore, performance chasing and lack of performance persistence are not contradictory.
- Active managers can be skilled even though investors earn zero expected excess returns.
- Rents from skill accrue to managers through fees and scale, not to investors through persistent alpha.
- The model predicts stronger flow-performance sensitivity for younger funds.
- Very poor performance can trigger fund exit when the posterior belief falls below the minimum viable threshold.
- The calibration can reproduce survival rates and the empirical flow-performance curve using a parsimonious parameterization.
- The model implies a high distribution of managerial skill, which contrasts with the common view that zero average alpha implies no skill.

8 Conclusion

8.1 Core conclusion

Berk and Green show that several empirical facts in the mutual fund literature are mutually consistent in a rational market. Investors rationally chase performance because performance is informative about skill. Performance is not persistent because fund inflows scale up successful managers and eliminate expected investor alpha. Active funds do not outperform benchmarks on average because capital is competitively supplied.

8.2 Economic intuition

The key economic intuition is that mutual fund markets adjust by quantity, not by security price. When a corporate project has positive expected NPV, the price of claims adjusts or firms raise capital on competitive terms. In an open-end mutual fund, the share price is tied to net asset value. The adjustment comes through fund flows. A successful manager receives more capital until diminishing returns make future investor excess returns zero.

8.3 Incremental lesson for empirical interpretation

The paper warns against equating zero investor alpha with zero managerial skill. In competitive markets, investors may earn zero surplus even when the underlying productive input is valuable. The surplus can be captured by the scarce input, here managerial talent. Thus, empirical tests of fund performance should distinguish investor returns from manager rents and from social value created by active management.

8.4 What the paper changes in the mutual fund debate

The model changes the interpretation of three facts.

- **Performance chasing** becomes rational learning and efficient capital allocation.
- **No persistence** becomes the equilibrium consequence of flows, not evidence against skill.
- **Zero average alpha** becomes evidence of competitive capital supply, not evidence that managers create no value.

8.5 Limitations and open questions

The model abstracts from search costs, distribution channels, marketing, fund families, taxes, heterogeneous investor liquidity needs, loads, agency problems, risk-taking incentives, and nonlinear fee contracts. These forces may matter in real mutual fund markets. But the paper's point is that one should not invoke those forces to explain facts that already follow from a rational benchmark with learning and decreasing returns.

8.6 Takeaway

The lasting takeaway is that mutual fund flows are an equilibrating mechanism. Capital flows toward managers whose histories reveal greater skill, and those flows eliminate future alpha for investors. This is why rational performance chasing, skilled managers, and unpredictable mutual fund returns can coexist.